(not so) Clean model description

October 21, 2026

1 Model description

Simplified Model

Jobs arrive over time and must be routed to one of two back-end servers:

- **Fast server:** faster processing, but incurs an additional routing cost.
- **Standard server:** slower processing, but no routing cost.

The fast server is not always the best choice because it may already have a long queue. Therefore, the routing decision trades off the lower processing time of the fast server against its routing cost and current congestion.

Each arriving job has two characteristics:

- a *job value* $v \in [0, 1]$, which determines the revenue generated by completing the job;
- a hidden binary state indicating whether the job is *simple* or *complex*.

Simple jobs require relatively little processing regardless of the selected server. In contrast, complex jobs require substantially more processing time, especially when routed to the standard server. Consequently, complex jobs benefit the most from being assigned to the fast server.

The decision maker does not observe whether a job is complex. However, by probing the job, an imperfect signal can be obtained. This signal is informative, but not perfectly accurate, and is used to improve the routing decision.

We consider a family of threshold policies of the form

$$\text{score} = a_v\, v - a_q\, (q_f - q_s) + a_s\, s,$$

where q_f and q_s denote the current workloads (or queue lengths) of the fast and standard servers, respectively, and $s \in \{0, 1\}$ is the probing signal. The job is routed to the fast server whenever

$$\text{score} \geq \tau,$$

where τ is the policy threshold.

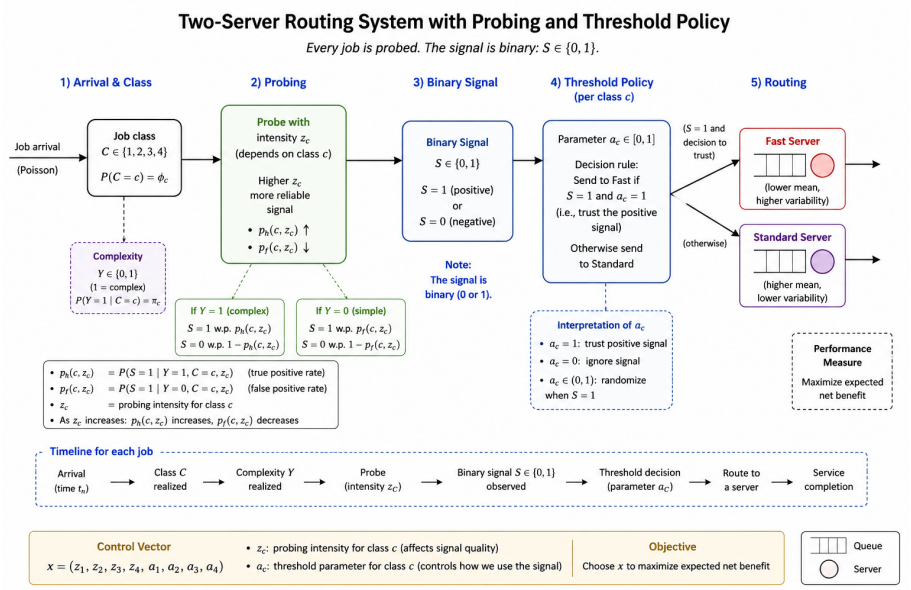


Figure 1: A summary of what is going on!

Example. Suppose

$$a_v = 0.7, \qquad a_q = 0.4, \qquad a_s = 0.8, \qquad \tau = 0.3.$$

Consider a job with value $v = 1$, signal $s = 1$, and equal queue lengths ($q_f = q_s$). The score is

$$0.7(1) - 0.4(0) + 0.8(1) = 1.5 > 0.3,$$

so the policy routes the job to the fast server. On the other hand, if the fast queue becomes substantially longer than the standard queue, the score decreases and the same policy may instead route the job to the standard server.

Job classes and complexity. Jobs belong to one of four classes. Each class has a different probability of being intrinsically complex:

$$\mathbf{p}^{\text{complex}} = (0.02,\ 0.12,\ 0.55,\ 0.95).$$

For example, only 2% of class-1 jobs are complex, whereas 95% of class-4 jobs are complex.

Conditional on the true complexity of the job, the probe generates an imperfect binary signal. Its quality is characterized by two vectors:

$$\mathbf{p}^{\text{hit}} = (0.25,\ 0.40,\ 0.75,\ 0.92), \qquad \mathbf{p}^{\text{false}} = (0.05,\ 0.08,\ 0.15,\ 0.20).$$

The first vector gives the probability that the probe correctly emits a positive signal when the job is complex (true positive), whereas the second gives the

probability that the probe incorrectly emits a positive signal when the job is simple (false positive).

Example. Consider a class-1 job. It is complex with probability 0.02. If it is indeed complex, then the probe correctly emits a positive signal with probability 0.25 (and misses it with probability 0.75). On the other hand, if the job is simple, the probe emits a false positive signal with probability 0.05. [de: Is there any concern with how p_1^{hit} and p_2^{hit} are both less than 0.50?] [bp: Yeah, they are maybe too low, meaning not worth paying for those. We can always change that.]

Probing effort. The decision variable $z_c \in [0, 1]$ represents the probing effort allocated to class-c jobs. This effort is fractional because probing is interpreted as an intensity: a larger value of z_c means that class-c jobs are inspected more carefully, using more time, resources, or diagnostic checks.

Probing improves the quality of the signal. For a class-c job, the true-positive and false-positive probabilities become

$$p_c^{\text{hit}}(z_c) = p_c^{\text{hit}} + (1 - p_c^{\text{hit}}) \left(1 - e^{-\beta z_c}\right),$$

and

$$p_c^{\text{false}}(z_c) = p_c^{\text{false}} e^{-\beta z_c}.$$

Thus, more probing increases the probability of detecting a complex job and decreases the probability of falsely flagging a simple job.

Example. For class 1, suppose

$$p_1^{\text{hit}} = 0.25, \qquad p_1^{\text{false}} = 0.05, \qquad \beta = 3, \qquad z_1 = 0.5.$$

Then

$$p_1^{\text{hit}}(0.5) = 0.25 + 0.75(1 - e^{-1.5}) \approx 0.833,$$

whereas

$$p_1^{\text{false}}(0.5) = 0.05 e^{-1.5} \approx 0.011.$$

Therefore, moderate probing substantially increases the probability of detecting a complex class-1 job, while also reducing false positives.

2 What could we improve/add to the model?

1. Maybe the routing policy should observe job class.
2. We should improve the story behind the classes. Something like
   (a) Class 1: routine requests,
   (b) Class 2: moderately difficult,

(c) Class 3: specialized,

(d) Class 4: highly customized.

3. The role of job value is still a bit weak. Right now, it only changes the revenue. It does not affect processing, due dates, or penalties. Consequently, it only influences routing indirectly. We might have a model in which higher-value jobs should also have tighter service requirements or higher tardiness penalties

3 Simulation Procedure

For a fixed probing decision z, the objective function can be interpreted as the following simulation procedure.

1. Generate one realization of the underlying system, consisting of a complete sample path
$$\xi_\ell = (\xi_{\ell 1}, \ldots, \xi_{\ell R}),$$
where each $\xi_{\ell r}$, $r = 1, 2, \ldots, R$, represents one simulated job in the queue and ℓ represents the replication index. This ℓth sample path is generated independently of the probing decision. [de: I took this last point to mean that we initially generate ξ_ℓ unconditionally (i.e., assuming no probing), but please correct me.]

2. Given the probing decision z, generate one realization of the probing outcomes, denoted by η_z. In the present model, η_z contains the probing results for all jobs and is a length-R vector.

3. Repeat the above operation n times, for a total of n independent replications of the simulation model.

4. [de: I paused revising this text/notation here because of my attempts to unify the setups (see below).] For each routing system $i = 1, \ldots, k$, estimate its conditional expected performance given the observed probing outcomes by
$$\frac{1}{n}\sum_{\ell=1}^{n} Y(x_i, \xi_\ell)\frac{f(\xi_\ell; \eta_z)}{f(\xi_\ell)}.$$
The likelihood ratio transforms the unconditional sample paths into an estimator of the conditional expectation given η_z.

5. Select the routing system with the largest conditional expected value,
$$\max_{i=1,\ldots,k} \frac{1}{n}\sum_{\ell=1}^{n} Y(x_i, \xi_\ell)\frac{f(\xi_\ell; \eta_z)}{f(\xi_\ell)},$$
and record this quantity as one realization of the outer expectation.

6. Repeat the previous steps for independent realizations of η_z and average the resulting values to estimate the objective function.

The likelihood-ratio approach exploits the fact that the underlying randomness of the system (arrival process, true job characteristics, service times, etc.) is independent of the probing decision z. Changing z only modifies the distribution of the probing outcomes and, consequently, the information available to the routing systems. As a result, the routing decisions and the realized system trajectory change with z, even though they are driven by the same underlying realization. This separation allows the underlying realizations to be recycled while accounting for different probing decisions through likelihood-ratio weights.

Saying the same thing again:

1. Nature generates the 200-job day.
2. z changes what the controller gets to observe.
3. The routing systems react differently because they receive different signals.
4. Therefore, the realized queues differ, despite sharing the same underlying randomness.

Could we have done things differently? Yeah, probably. Maybe probing would be a marketing effort that would increase the average number of jobs and even their value.

4 Is that what is in the mysterious Dr. Eckman's mind?

Since the probing decision z does not modify the underlying stochastic process (arrival process, true job characteristics, service times, etc.), the unconditional distribution of the sample paths remains unchanged. The only distribution affected by z is that of the probing outcomes.

Consequently, Bayes' theorem yields

$$f(\xi \mid \eta, z) = \frac{f(\xi)\, f(\eta \mid \xi, z)}{f_z(\eta)},$$

or equivalently,

$$\frac{f(\xi \mid \eta, z)}{f(\xi)} = \frac{f(\eta \mid \xi, z)}{f_z(\eta)}.$$

Substituting this expression into our famous equation(7) gives

$$J(z) = -\alpha^\top z + \int_{\mathbb{R}^{d_z}} \left[\max_{i=1,\dots,k} \frac{1}{n} \sum_{\ell=1}^{n} Y(x_i, \xi_\ell) \frac{f(\eta_z \mid \xi_\ell)}{f_z(\eta_z)} \right] f_z(\eta_z)\, d\eta_z.$$

Since $f_z(\eta_z)$ is independent of the routing system i, it follows that

$$f_z(\eta_z) \max_i \left\{ \frac{A_i(\eta_z)}{f_z(\eta_z)} \right\} = \max_i A_i(\eta_z),$$

and therefore

$$\boxed{J(z) = -\alpha^\top z + \int_{\mathbb{R}^{d_z}} \max_{i=1,\ldots,k} \left[\frac{1}{n} \sum_{\ell=1}^{n} Y(x_i, \xi_\ell) f(\eta_z \mid \xi_\ell) \right] d\eta_z.}$$

I like this derivation because it isolates exactly where the simplification comes from: not from the likelihood ratio itself, but from the fact that $f(\xi)$ is independent of z. The marginal density $f_x(\eta)$ appears naturally through Bayes' rule and then cancels against the outer expectation. This is a much cleaner argument than simply asserting that the likelihood ratio simplifies the expression.

[de: The name of that last section really cracked me up.

I see things a little differently, as I'll try to describe here. It may be that things end up at the same place as the analysis above, but by taking a different path.]

[bp: Believe me, Franco and I asked ourselves a few times "what does he mean by simplifying the expression???"]

Let ξ represent the aleatoric uncertainty in a single replication. In our example, ξ consists of the arrival times, service times (both in the first stage and the two possibilities in the second stage—at the fast and at the standard server, even though only one of these values will be used), the classes of the jobs, the complexities of the jobs, and the signals (probing outcomes) for all jobs. IMPORTANT: The signal MUST be part of ξ because the routing decision (the simulation model's logic) depends on the signal.

In terms of our notation, η_z is the list of probing outcomes/signals (one for each job) *given a probing decision* z. Hence, one component of ξ equals η_z. [de: I believe you mentioned one of the first papers on probing making such an assumption. The fact that we have this degeneracy in our problem somewhat distorts the derivations of the likelihood ratios and gradient estimators; see what is to come. I think that is find for an initial example, but I believe the setup we envisage is the more general one, where η does not directly enter into the simulation model's logic.] [bp: I see, I see. So. you mean that the general case would be something like

$$\xi = (\lambda_z, \mu_z, \theta_z, \nu_z, \eta_z),$$

right???] [de: Yes. There may be some advantage in defining $\xi = (\xi_{-z}, \xi_z)$ where the portion ξ_z depends on z and the portion ξ_{-z} does not. In our example, $\xi_{-z} = (\lambda, \mu, \theta, \nu)$ while $\xi_z = \eta_z$.]

[bp: How would things change if we had, say, $\xi_z = (\lambda_z, \eta_z)$? Just a harder gradient computation? Maybe having to use simulation for it?]

[bp: It seems the first big change (what we called level II in our discussions) would be z generating a distribution on the values of ξ_z, which would require the famous external integral of the even most famous equation (7). It seems messier, but essentially the same logic.]

[bp: The different setting would really be z changing the system. In summary, I see things like that now:

Distributional probing:	$z \longrightarrow \xi_z$	$\xi_z \sim P(\cdot; z)$,
Random parameter	$z \longrightarrow \theta_z \longrightarrow \xi$	$\theta_z \sim P_\theta(\cdot; z), \quad \xi \sim P_\xi(\cdot \mid \theta_z)$,
Configuration/design:	$z \longrightarrow \mathcal{C}(z)$	$Y = Y(x, \xi; \mathcal{C}(z))$.

]

[de: Note: For the next part, we will be talking about a single replication and will suppress the replication index ℓ from the subscript on ξ. It will come back later when we write the overall LR estimator.] Assume, for simplicity, that the stochastic elements are independent across jobs, i.e., $\xi_1, \xi_2, \ldots, \xi_R$ are independent. Within a given ξ_r (job r), however, the stochastic elements are not independent. For example, the complexity of the job depends on the problem class, and the probing outcome (signal) depends on the complexity and the job class. The service times also depend on the complexities and the job classes, but this is less important.

Let $f(\xi)$ represent the *unconditional* density of the stochastic primitives, assuming NO PROBING. The probing outcomes therefore follow their default distributions when $z = \mathbf{0}$. As a result of the assumption of independence across jobs, the density $f(\xi)$ decomposes into a product of densities: $f(\xi_1) \times f(\xi_2) \times \cdots \times f(\xi_R)$. In what follows, we focus on one such $f(\xi_r)$.

$$
\begin{aligned}
f(\xi_r) &= f(\text{arrival, services, complexity, class, signal}) \\
&= f(\text{arrival, services} \mid \text{complexity, class}) \times f(\text{signal} \mid \text{complexity, class}) \\
&\quad \times f(\text{complexity} \mid \text{class}) \times f(\text{class})
\end{aligned}
$$

Of the four terms in $f(\xi_r)$, signal appears only in the second, $f(\text{signal} \mid \text{complexity, class})$. By defining signal $= \eta_{z,r}$, we also have that

$$
\begin{aligned}
f(\xi_r; \eta_{z,r}) = f(&\text{arrivals, services} \mid \text{complexity, class}) \times 1 \\
&\times f(\text{complexity} \mid \text{class}) \times f(\text{class}).
\end{aligned}
$$

The (incomplete, as you will see) likelihood ratio is then

$$
\frac{f(\xi; \eta_z)}{f(\xi)} = \prod_{r=1}^{R} \frac{f(\xi_i; \eta_{z,r})}{f(\xi_r)} = \prod_{r=1}^{R} \frac{1}{f(\text{signal} \mid \text{complexity, class}, z = \mathbf{0})}.
$$

We further need to incorporate $f(\eta_{z,r}; z)$ into the numerator of the likelihood ratio. This too will be *conditional on the complexity and class*, which is not

clearly represented in the notation at this point. [de: My current thinking is that since $\eta_{z,r}$ appears in ξ_r, the likelihood ratio we really want should condition on z rather than η_z (see the upcoming analysis).] What this will lead to is a likelihood ratio of the form

$$\frac{f(\xi; z)}{f(\xi)} = \prod_{r=1}^{R} \frac{f(\text{signal} \mid \text{complexity, class}, z)}{f(\text{signal} \mid \text{complexity, class}, z = \mathbf{0})}.$$

Equation (7) should then look like

$$\boxed{J(z) = -\alpha^\top z + \max_{i=1,\ldots,k} \left[\frac{1}{n} \sum_{\ell=1}^{n} Y(x_i, \xi_\ell) \frac{f(\xi_\ell; z)}{f(\xi_\ell)} \right].}$$

[de: Some new notation.] For job $r = 1, 2, \ldots, R$, let C_r be its class, D_r be its complexity, S_r be its signal. Here, the upper-case letters will denote the *random* interpretations of these quantities and the lower-case letters (c_r, d_r, s_r), will denote the realized values. The realized values will be needed later in the likelihood ratios.

The LR in this problem is

$$\begin{aligned}
\frac{f(\xi_\ell; z)}{f(\xi_\ell)} &= \prod_{r=1}^{R} \frac{P(S_r = s_r | D_r = d_r, C_r = c_r; z)}{P(S_r = s_r | D_r = d_r, C_r = c_r; \text{no probing})} \\
&= \prod_{r:\ \substack{s_r=\text{positive} \\ d_r=\text{complex}}}^{R} \frac{p^{\text{hit}}_{c_r}(z_{c_r})}{p^{\text{hit}}_{c_r}(0)} \prod_{r:\ \substack{s_r=\text{negative} \\ d_r=\text{complex}}}^{R} \frac{1 - p^{\text{hit}}_{c_r}(z_{c_r})}{1 - p^{\text{hit}}_{c_r}(0)} \\
&\quad \times \prod_{r:\ \substack{s_r=\text{positive} \\ d_r=\text{simple}}}^{R} \frac{p^{\text{false}}_{c_r}(z_{c_r})}{p^{\text{false}}_{c_r}(0)} \prod_{r:\ \substack{s_r=\text{negative} \\ d_r=\text{simple}}}^{R} \frac{1 - p^{\text{false}}_{c_r}(z_{c_r})}{1 - p^{\text{false}}_{c_r}(0)}
\end{aligned}$$

[de: REGARDE ICI: The blue LR was previously missing from the expression for $\nabla_z J(z)$ below. I am fairly certain it needs to be there, because otherwise the red term alone has no dependence on the z under which the ξ's were simulated, which does not make sense.] The associated LR gradient estimator is

$$\boxed{\nabla_z J(z) = -\alpha + \frac{1}{n} \sum_{\ell=1}^{n} Y(x_{i^*}, \xi_\ell) \frac{f(\xi_\ell; z)}{f(\xi_\ell)} \nabla_z \log f(\xi_\ell; z)}$$

where

$$i^* = \arg \max_{i=1,\ldots,k} \left[\frac{1}{n} \sum_{\ell=1}^{n} Y(x_i, \xi_\ell) \frac{f(\xi_\ell; z)}{f(\xi_\ell)} \right].$$

The score function term in this problem is

$$\begin{aligned}
\nabla_z \log f(\xi_\ell; z) &= \sum_{r=1}^{R} \nabla_z \log P(S_r = s_r | D_r = d_r, C_r = c_r; z) \\
&= \sum_{\substack{r:\ s_r=\text{positive} \\ d_r=\text{complex}}}^{R} \nabla_z \log p_{c_r}^{\text{hit}}(z_{c_r}) + \sum_{\substack{r:\ s_r=\text{negative} \\ d_r=\text{complex}}}^{R} \nabla_z \log \left(1 - p_{c_r}^{\text{hit}}(z_{c_r})\right) \\
&\quad + \sum_{\substack{r:\ s_r=\text{positive} \\ d_r=\text{simple}}}^{R} \nabla_z \log p_{c_r}^{\text{false}}(z_{c_r}) + \sum_{\substack{r:\ s_r=\text{negative} \\ d_r=\text{simple}}}^{R} \nabla_z \log \left(1 - p_{c_r}^{\text{false}}(z_{c_r})\right) \\
&= \sum_{\substack{r:\ s_r=\text{positive} \\ d_r=\text{complex}}}^{R} \frac{(1 - p_{c_r}^{\text{hit}}(0))\beta e^{-\beta z_{c_r}}}{p_{c_r}^{\text{hit}}(z_{c_r})} + \sum_{\substack{r:\ s_r=\text{negative} \\ d_r=\text{complex}}}^{R} \frac{-(1 - p_{c_r}^{\text{hit}}(0))\beta e^{-\beta z_{c_r}}}{1 - p_{c_r}^{\text{hit}}(z_{c_r})} \\
&\quad + \sum_{\substack{r:\ s_r=\text{positive} \\ d_r=\text{simple}}}^{R} \frac{-p_{c_r}^{\text{false}}(0)\beta e^{-\beta z_{c_r}}}{p_{c_r}^{\text{false}}(z_{c_r})} + \sum_{\substack{r:\ s_r=\text{negative} \\ d_r=\text{simple}}}^{R} \frac{p_{c_r}^{\text{false}}(0)\beta e^{-\beta z_{c_r}}}{1 - p_{c_r}^{\text{false}}(z_{c_r})}
\end{aligned}$$

[de: New text about the green gradient search.]

The LR gradient estimator $\nabla_z J(z)$ derived above can be evaluated at any z. Hence, we can think of $J(z)$ as a surrogate for the objective function, which we can then attempt to optimize using a gradient based search. In the setup we have tested, we perform n initial replications under the no-probing case and obtain a static surrogate J that we then optimize. We know J will be biased and (possibly) highly variable for some z, like those z far from 0. As we increase n, we expect the bias and variable to both vanish, but possibly at a slow rate for z far from 0.

We could alternatively take n observations at each iterate z^τ ($\tau = 0, 1, 2, \dots$) visited by the search and use only these n observations to compute a LR gradient estimate and thereby take a gradient step. The advantage of this approach is that LR gradient estimator should be more accurate because it is based on replications under the current z (hence the likelihood ratio in $\nabla_z J(z)$ is uniformly 1 and therefore vanishes). The downside of this approach is that it essentially discards all previous simulation replications taken under past probing decisions.

The idea behind green simulation is to achieve a greater effective sample size (and therefore a less variable LR gradient estimator) by reuse historical observations previously collected throughout the search. These observations are appropriately reweighted by a LR and score function, both of which depend on the current iterate: the z^τ where the gradient is being estimated. [de: Aside: An early paper of mine on the green gradient algorithm demonstrated that there is an additional bias introduced by this reuse due to the observations from different iterates being *dependent*. This dependence arises because the search trajectory (and hence the observations you will see next) depends on the observations you already saw.]

Suppose we have run a gradient search in which we have visited a sequence of probing decisions $z^0, z^1, \ldots, z^t$ and have generated n i.i.d. observations at each of these iterates. The green LR gradient estimator for the gradient at z^t is

$$\boxed{\nabla_z J^{\text{green}}(z^t) = -\alpha + \frac{1}{t+1}\sum_{\tau=0}^{t}\left[\frac{1}{n}\sum_{\ell=1}^{n} Y(x_{i^*}, \xi_\ell; z^\tau)\frac{f(\xi_\ell; z^t)}{f(\xi_\ell; z^\tau)}\nabla_z \log f(\xi_\ell; z^t)\right]}$$

where the z^τ in the notation $Y(x_{i^*}, \xi_\ell; z^\tau)$ reflects that this observation was obtained when ξ_ℓ was generated under the conditional distribution associated with z_τ, i.e., it's one of the n outputs obtained when running the simulation model under a probing decision z^τ previously visited during the gradient search. The score function $\nabla_z \log f(\xi_\ell; z)$ is the same as that shown above, and the likelihood ratio $\frac{f(\xi_\ell; z^t)}{f(\xi_\ell; z^\tau)}$ is nearly the same as that shown above, except with $p_{c_r}^{\text{hit}}(z_{c_r}^\tau)$ and $p_{c_r}^{\text{false}}(z_{c_r}^\tau)$ in the denominators instead of $p_{c_r}^{\text{hit}}(0)$ and $p_{c_r}^{\text{false}}(0)$.

5 Notes on Differences Compared to Equation (7)

[de: This text is new.] When we first derived Equation (7), the original problem was

$$\max_{z\in[0,1]^m}\left\{-\alpha^T z + \mathbb{E}_\eta\left[\max_{i=1,2,\ldots,k}\mathbb{E}\left[Y(x_i,\xi)\mid \eta; z\right]\right]\right\}.$$

In this setup, we were able to first observe η (from probing) and then decide on an optimal ACTION i before experiencing a reward Y that depended on further randomness ξ. In our production example, on the other hand, we have to commit to a POLICY beforehand when we choose the coefficients and thresholds. This policy then dictates which action we take when we see a certain probing signal η. In the production example, i corresponds to the policy parameters, not the decision to route the job to the standard or fast line. The examples we've been looking at recently do not have the full "two-stage" setup where we pick z, observe η, pick i, and observe ξ. Instead, we are simultaneously picking z and i and then experiencing ξ (which now includes η). Put differently, there is no expectation outside the maximum, hence our formulas for the LR estimators did not involve the Monte Carlo integration that is present in the more general two-stage model we started with.

6 Example 2: Predictive Maintenance with Continuous Inspection Effort

[de: My two cents: I like the story told by this example. Other than the use of normal distributions instead of Bernoulli's in the likelihood ratios, I'm not sure it's fundamentally different from the production example. Still, I really like it.]

[bp: Well, I did not have time to explain why I came up with this example. I FULLY agree it is quite similar to the other example. My goal was to dig deeper into the story of what we are doing. Before coming with this example I tried a lot of other ideas that did not work out. I tried to adapt some well-known problems in the POMDP literature in which you have information acquisition/learning, but they felt flat. The trade-off was just not there, or at least I could not see it. This one seems to be a better blueprint for what we want to do.]

Consider a system consisting of machines that are subject to degradation and unexpected failures. At an inspection epoch, each machine has an observable age $A \in [0,1]$ and an unobservable degradation level $D \in [0,1]$. Older machines tend to be more degraded, but age alone does not perfectly reveal the true condition of a machine. We model degradation as

$$D = [0.10 + 0.72A + 0.16Z_D]_{[0,1]}, \qquad Z_D \sim N(0,1),$$

where $[\cdot]_{[0,1]}$ denotes truncation to the interval $[0,1]$.

The decision maker can acquire additional information about the condition of the machine through an inspection. Inspection produces a noisy signal

$$S = D + \varepsilon,$$

where the precision of the signal depends on the inspection effort.

We consider two age groups,

$$g(A) = \begin{cases} 1, & A < 0.55, \\ 2, & A \geq 0.55, \end{cases}$$

corresponding to younger and older machines, respectively. The probing decision is

$$z = (z_1, z_2) \in [0,1]^2,$$

where z_1 is the inspection effort devoted to younger machines and z_2 is the inspection effort devoted to older machines.

Conditional on the true degradation level D, the inspection signal satisfies

$$S \mid D, z_g \sim N\left(D, \sigma_g^2(z_g)\right),$$

where

$$\sigma_g(z_g) = \sigma_{0,g} \exp(-\beta_g z_g).$$

for $g = 1, 2$. Thus, increasing z_g reduces the variance of the inspection error and provides more accurate information about the true condition of the machine. Inspection is costly, with total information-acquisition cost

$$C(z) = \alpha_1 z_1 + \alpha_2 z_2.$$

After observing the machine's age A and inspection signal S, the decision maker chooses whether to perform preventive maintenance or to keep the machine in operation. We consider a family of threshold policies of the form

$$\pi_\theta(A, S) = \begin{cases} \text{preventive maintenance}, & a_S S + a_A A \geq \tau, \\ \text{continue operation}, & a_S S + a_A A < \tau, \end{cases}$$

where

$$\theta = (a_S, a_A, \tau)$$

denotes the parameters of the maintenance policy.

The threshold policy combines two sources of information. Age A is observed freely and provides prior information about machine condition, whereas S is costly information whose accuracy depends on the probing decision z.

If preventive maintenance is performed, a moderate and predictable maintenance cost is incurred. In the numerical experiment, this cost is approximately

$$C_{\mathrm{PM}} = 18.$$

Preventive maintenance may therefore be undesirable when performed too early, since a machine with substantial remaining useful life is replaced or repaired unnecessarily.

If preventive maintenance is not performed, the machine continues operating and may experience an unexpected failure. Conditional on age and degradation, the failure hazard is modeled as

$$\lambda(A, D) = \lambda_0 \exp\left(\gamma_D D + \gamma_A A\right).$$

A potential failure time is generated according to

$$T_F \mid A, D \sim \text{Exponential}\big(\lambda(A, D)\big).$$

If T_F occurs within the operating horizon, the machine fails unexpectedly. Such a failure is much more expensive than preventive maintenance, with a corrective-maintenance cost of approximately

$$C_{\mathrm{fail}} = 105,$$

in addition to production losses associated with unplanned downtime.

The fundamental trade-off is therefore

premature preventive maintenance	versus	costly unexpected failure.

Inspection information is valuable because it helps distinguish machines that should be maintained from machines that can safely remain in operation. However, more accurate information requires greater inspection expenditure. The outer optimization problem therefore determines how much information should be acquired:

$$\max_{z \in [0,1]^2} \left\{ \max_{\theta \in \Theta} \mathbb{E}\left[Y(\theta, \xi) \mid z\right] - \alpha^\top z \right\},$$

where Θ is the finite set of candidate threshold policies and ξ contains the random machine characteristics, inspection signal, and failure time.

[de: For "random machine characteristics," I assume this refers to the machine's age and degradation. Because age is observable, we may want to treat it as deterministic. As for the degradation, to run a replication, we would need

a probability model for the distribution of D, for the sake of generating random machine characteristics. But where would this model come from? It would seem to correspond to the no-probing case ($z = 0$). The inspection signal and failure time both correspond to aleatoric uncertainty (risk), whereas the random degradation is more like our epistemic uncertainty (uncertainty about the state of the world). [In the production example, we knew the underlying probability of jobs of certain classes being simple or complex and probing only told us something about the individual jobs.] In this example, only the inspection signal depends on probing, so again S is the natural choice for the η in our notation. Like in the production example, the simulation code needs the signal, so S is also part of ξ and is fully observed given a probing decision. Long story short, I think this example has the same setup as the production example, where the outer integral goes away.]

[bp: Let's see, let me clarify all that. First, $\xi_z = S, \xi_{-z} = (A, D, T_F)$. As I said before, I fully agree this belong to the same level (I), same spirit, same setup, same category of the factory example. Zero disgreement here. Regarding Age, I think it could be deterministic, or we could generate ages randomly. I see this more as a modelling choice.

Yes, D requires an underlying degradation model. My interpretation is that

$$D \mid A \sim P_D(\cdot \mid A)$$

represents the baseline distribution of the true physical degradation among machines of a given observed age. In a real application, this distribution would be estimated or calibrated from historical reliability, inspection, or condition-monitoring data.

Importantly, this is not specifically a "no-probing" ($z = 0$) distribution. The machine has a realized true degradation state D regardless of whether we inspect it. The probing decision does not generate or modify D; it only determines how accurately we observe its realized value through the inspection signal:

$$D \mid A \sim P_D(\cdot \mid A), \qquad S \mid D, z \sim P_S(\cdot \mid D, z).$$

Thus, D belongs to the baseline uncertainty of the physical system and is independent of z, whereas S is the z-dependent random variable. In our numerical example, we simply posit a synthetic model for $D \mid A$ for simulation purposes. I agree that S therefore plays the same role as the probing signal in the production example, and the example has the same basic structure with respect to the probing decision.

]

[de: Some details of what constitutes a replication are still unclear to me. Is there some finite time horizon and a fixed set of machines and we simultaneously make decisions to repair/not repair each machine and then watch them break down (or not) and compute a cost.]

[bp: Yeah, well, this was mostly a quick placeholder. Details are in the code! Here is a more detailed explanation. If you like it, feel free to add to the body of the text: A replication is simpler than that. In the current implementation, one

replication corresponds to one machine at one inspection epoch. The machine has an observed age A and a latent degradation state D. Given the probing decision z, an inspection signal S is generated, and the threshold policy then decides whether to perform preventive maintenance or to continue operating the machine. If preventive maintenance is performed, the corresponding preventive maintenance cost is incurred and the replication ends. Otherwise, the machine is operated over a fixed finite horizon (normalized to one time unit). A potential failure time T_F is generated from a distribution whose hazard depends on the machine's age and true degradation. If T_F falls within the operating horizon, an unexpected failure occurs and the corresponding failure and downtime costs are incurred; otherwise, the machine survives the horizon.

Thus, one replication can be summarized as

$$(A, D) \longrightarrow S(z) \longrightarrow \{\text{preventive maintenance or continue}\} \longrightarrow \{\text{survive or fail over the operating horizon}\}.$$

The simulation then averages the cost over many independent machine episodes. There is currently no fixed fleet of interacting machines, no maintenance capacity constraint, and no queue for maintenance.]

[de: This may be a crazy idea, but it may be possible to extend the model to a continuous monitoring setting. Please bear with me... Age would increase linearly with time, i.e., $A(t) = A_0 + t$ where A_0 is the age of the machine at the start of the time horizon. The signal would be continuously monitored where $\varepsilon(z_{g(A)})$ is now a drift-zero Brownian motion with diffusion $\sigma_g^2(z_g)$. (A nuisance with this is that the machine's group g could change over time, which changes the diffusion discontinuously, but whatever.) We would also need another model for how the degradation increases over time (with greater use) and how the machine fails; the current failure hazard model assumes the age and degradation of the machine are fixed, whereas we would now have both increase over time. This is all stream-of-conscious, so please just ignore this for now.]

[bp: Ha! I like it! If you read my previous comments (I wrote things in order without verifying what you wrote after) you sort of fulfilled my prophecy of "it makes sense to play around a bit more and propose new examples of the type 1. Nice ideas could come out of that." I will try to have something VERY RUDIMENTARY based on your example for the meeting.]

The sequence of events for a machine can be summarized as

$$\boxed{\begin{aligned}(A, D) \;&\longrightarrow\; z_{g(A)} \;\longrightarrow\; S = D + \varepsilon(z_{g(A)}) \text{ where } \varepsilon(z_{g(A)}) \sim N(0, \sigma_g^2(z_g)) \\ &\longrightarrow\; \pi_\theta(A, S) \;\longrightarrow\; \begin{cases}\text{preventive maintenance,} \\ \text{continued operation and possible failure.}\end{cases}\end{aligned}}$$

The role of continuous probing is particularly transparent in this example. At low inspection effort, the signal S is noisy, so maintenance decisions rely heavily on age and may result either in unnecessary preventive maintenance or missed detection of highly degraded machines. Increasing inspection effort

improves the accuracy of S, allowing a better maintenance policy to be implemented. Beyond some point, however, the marginal value of additional precision may be smaller than its marginal inspection cost, potentially leading to an interior optimal probing decision z^*.

7 Example 3: the well-known Eckman maintenance model

Here are some very preliminary results for the well-known Eckman continuous maintenance problem:

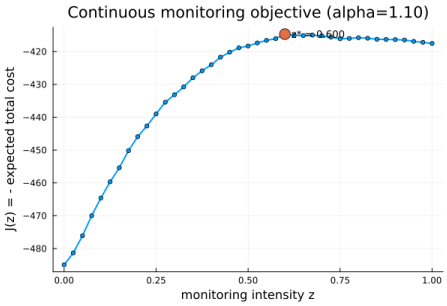


Figure 2: (a) Objective function as a function of monitoring intensity z.

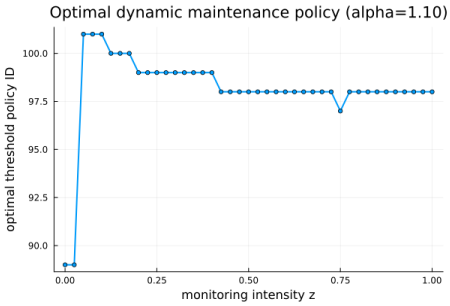


Figure 3: (b) Optimal dynamic maintenance policy as a function of monitoring intensity z.

Figure 4: Continuous-monitoring predictive maintenance. Increasing the monitoring intensity z improves the quality of information about machine degradation but incurs a higher monitoring cost. The resulting trade-off yields an interior optimal monitoring intensity of approximately $z^* = 0.60$.

8 Example 4: A possible level II case

8.1 Production and rework system

We consider a production system composed of multiple parallel assembly lines, a quality-control station, and a centralized rework center. Let $\mathcal{C} = \{1, \ldots, m\}$ denote the set of assembly lines or product families and denote the set of assembly lines, and let $[0, H]$ denote the operating shift. Each assembly line $c \in \mathcal{C}$ continuously produces units of a particular product family during an operating shift.

After assembly, every completed unit undergoes a quality-control test. A unit that passes the test is conforming and proceeds directly to packaging and shipment. A unit that fails the test requires corrective processing and becomes a

rework job. These defective units arrive dynamically at the centralized rework center as production progresses. The line of origin is observed when the job arrives at the rework center. However, the job may require either simple or complex rework. Its complexity is uncertain and is influenced probabilistically by the latent condition of its assembly line of origin.

The rework center contains two alternative resources:

- the fast server, representing an expert technician or an advanced repair station with shorter service times but a higher operating cost;
- the standard server, representing a regular repair station with a lower operating cost but longer service times, particularly for complex jobs.

Each assembly line $c \in \mathcal{C}$ is associated with a known product value $v_c \in [0, 1]$. Before the next job arrives, its value is unknown because its line of origin is unknown. Once the job arrives, however, its line of origin is observed and its value v_c is determined. Therefore, job values are not included as separate random elements in ξ.

After service at either server, the repaired unit is released back to the normal production flow and subsequently proceeds to packaging and shipment. Thus, conforming units bypass the rework center, while defective units are routed through one of the two rework resources before reaching the final production stage.

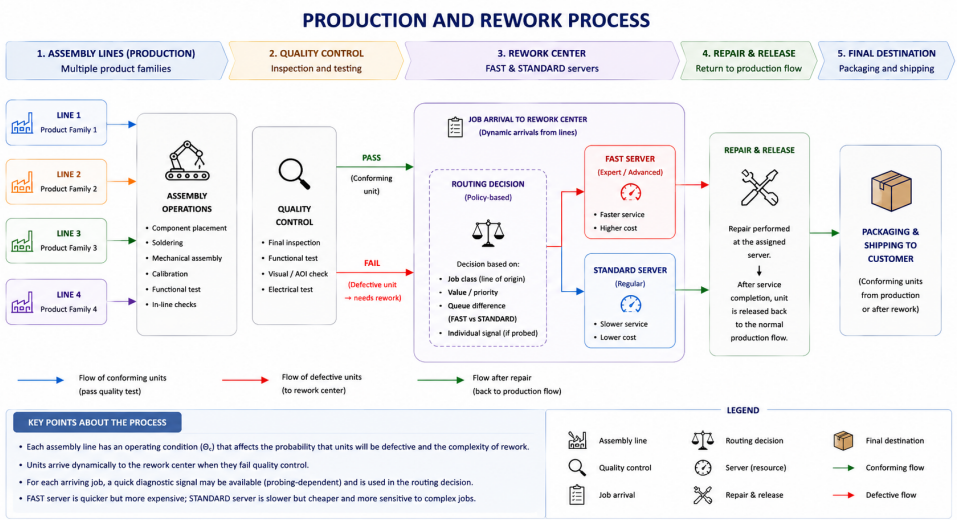


Figure 5: Assembly production system with rework

8.2 Condition of the assembly lines

The operating condition of assembly line c during a shift is represented by a latent variable

$$\Theta_c \in \mathbb{R}.$$

The variable Θ_c captures factors such as tool wear, misalignment, temperature instability, contamination, or equipment degradation. A larger value of Θ_c represents a more adverse operating condition. Let $\Theta = (\Theta_1, \ldots, \Theta_m)$ denote the vector of latent assembly-line conditions. Thus, the state of assembly lines Θ takes an unknown value before the probing decision and remains unknown and approximately constant throughout the operating shift. The diagnostic/inspection (probing) decision does not modify the physical condition of the line; it only provides information about it.

We assume that

$$\Theta_c \sim \mathcal{N}\left(\mu_c, \tau_c^2\right), \qquad c \in \mathcal{C},$$

and that the line conditions are mutually independent. The parameters μ_c and τ_c^2 represent the prior mean and variance of the operating condition of line c, respectively.

The condition of each assembly line affects its future rework stream through two mechanisms. First, it affects the rate at which rework jobs arrive at the centralized rework center. Second, the latent condition affects the probability that a rework job requires complex corrective processing.

We assume that the arrival rate is defined as

$$\lambda_c(\Theta_c) = \lambda_c^0 \exp\left[\gamma_c^\lambda \left(\Theta_c - \mu_c\right)\right], \qquad \gamma_c^\lambda \geq 0,$$

where $\lambda_c^0 > 0$ is the rework-arrival rate when $\Theta_c = \mu_c$, and γ_c^λ measures its sensitivity to the line condition.

For the jth rework job generated by line c, let

$$D_{c,j} = \begin{cases} 0, & \text{if the job requires simple rework,} \\ 1, & \text{if the job requires complex rework.} \end{cases}$$

Conditional on Θ_c, we assume that

$$D_{c,j} \mid \Theta_c \sim \text{Bernoulli}\left(p_c(\Theta_c)\right),$$

where

$$p_c(\Theta_c) = \frac{1}{1 + \exp\left[-\left(\gamma_{c,0}^p + \gamma_{c,1}^p \left(\Theta_c - \mu_c\right)\right)\right]}, \qquad \gamma_{c,1}^p \geq 0.$$

Here, $p_c(\Theta_c)$ denotes the probability that a rework job generated by line c requires complex processing. The parameter $\gamma_{c,0}^p$ is the baseline log-odds of complex rework when $\Theta_c = \mu_c$, whereas $\gamma_{c,1}^p$ controls how this probability changes with the line condition.

Consequently, an unfavorable line condition can generate more rework jobs and a larger proportion of complex jobs. [de: So larger values of Θ_c are more unfavorable.] [fq: Yes. Note that I changed Θ to be real-valued, as you correctly pointed out that Θ should be normally distributed.] Nevertheless, the future arrival times and the complexity requirements of the individual rework jobs remain uncertain.

8.3 Pre-shift information acquisition

Before the production shift begins, the decision maker may allocate diagnostic downtime to each assembly line to obtain information about its operating condition. Let

$$z_c \in [0, 1]$$

denote the probing effort allocated to line c. The variable z_c is interpreted as the fraction of the maximum available diagnostic time assigned to that line.

If $T_c^{\max}$ is the maximum diagnostic duration, the actual diagnostic time is

$$t_c(z_c) = T_c^{\max} z_c.$$

Thus, $z_c = 0$ means that no additional diagnostic inspection is performed, whereas $z_c = 1$ means that the maximum diagnostic time is used.

During the diagnostic period, inspection collect information about vibration, alignment, temperature, electrical stability, or other indicators of the line condition. For $z_c > 0$, the resulting measurement is modeled as

$$\eta_c \mid \Theta_c, z_c \sim \mathcal{N}\left(\Theta_c, \frac{\nu_c}{T_c^{\max} z_c}\right),$$

where $\nu_c > 0$ is a parameter controlling the intrinsic noise of the diagnostic procedure for line c. Equivalently,

$$\eta_c = \Theta_c + \varepsilon_c(z_c),$$

where

$$\varepsilon_c(z_c) \sim \mathcal{N}\left(0, \frac{\nu_c}{T_c^{\max} z_c}\right).$$

The measurement errors are assumed to be mutually independent and independent of the latent line conditions.

Longer diagnostic times produce more precise information:

$$z_c \uparrow \quad \Longrightarrow \quad \mathrm{Var}\left(\eta_c \mid \Theta_c, z_c\right) \downarrow .$$

When $z_c = 0$, no measurement is obtained for line c, and the decision maker retains the prior distribution of Θ_c.

The probing decision is costly because the assembly line must be stopped during the diagnostic procedure. Let g_c denote the expected production margin

lost per unit of downtime and let d_c denote the diagnostic cost per unit of time. The probing cost for line c is

$$C_c^{\text{probe}}(z_c) = (g_c + d_c)T_c^{\max} z_c = \alpha_c z_c.$$

The total probing cost is therefore

$$C^{\text{probe}}(z) = \sum_{c \in \mathcal{C}} \alpha_c z_c = \alpha^\top z.$$

The probing operation is assumed to occur immediately before the modeled operating horizon. Its effect on lost production is represented through $C^{\text{probe}}(z)$ rather than by explicitly modifying the arrival process during the subsequent shift.

8.4 Posterior predictions of rework arrivals and complexity

After observing the diagnostic measurements

$$\eta = (\eta_1, \ldots, \eta_m),$$

the decision maker updates the distribution of the condition of every measured assembly line:

$$\mathbb{P}\left(\Theta_c \mid \eta_c, z_c\right).$$

[de: Given η_c, it isn't clear to me why the posterior distribution would additionally depend on z_c. Could this instead be $\mathbb{P}(\Theta_c \mid \eta_c)$?] [fq: I think the posterior distribution should depends on both η_c and z_c because the probing decision determines the duration of the inspection, and hence the precision of the diagnostic measurement (by reducing the variance). In particular, as

$$\eta_c \mid \Theta_c, z_c \sim \mathcal{N}\left(\Theta_c, \frac{\sigma_c^2}{T_c^{max} z_c}\right),$$

then the likelihood of a given measurement depends on z_c. The same numerical value of η_c therefore provides different evidence about Θ_c depending on the diagnostic time used to obtain it.]

[de: Because Θ_c will be part of ξ in our analysis, we need to specify the form of the posterior distribution of Θ_c given η_c.]

[fq: Can it be something like this?

Given the prior distribution

$$\Theta_c \sim \mathcal{N}\left(\mu_c, \tau_c^2\right),$$

the posterior distribution is also normal. For $z_c > 0$,

$$\Theta_c \mid \eta_c, z_c \sim \mathcal{N}\left(m_c(\eta_c, z_c), v_c(z_c)\right),$$

where

$$v_c(z_c) = \left(\frac{1}{\tau_c^2} + \frac{T_c^{\max} z_c}{\nu_c} \right)^{-1}$$

and

$$m_c(\eta_c, z_c) = v_c(z_c) \left(\frac{\mu_c}{\tau_c^2} + \frac{T_c^{\max} z_c}{\nu_c} \eta_c \right).$$

When $z_c = 0$, no measurement is obtained and the posterior distribution coincides with the prior distribution.]

[de: Since the signal η is normally distributed with known variance, we have a normal likelihood. The corresponding conjugate prior for Θ_c would then be a normal distribution. But since Θ_c is bounded between $[0, 1]$, a normal prior/posterior for Θ_c may be a little unnatural.]

[fq: You are right. I modified Θ_c to be a real-valued latent condition index and assumed it to be normally distributed.]

The posterior distribution induces updated predictions for the future rework stream generated by each assembly line. In particular, define

$$\widehat{\lambda}_c(\eta_c, z_c) = \mathbb{E}\left[\lambda_c(\Theta_c) \mid \eta_c, z_c\right]$$

and

$$\widehat{p}_c(\eta_c, z_c) = \mathbb{E}\left[p_c(\Theta_c) \mid \eta_c, z_c\right].$$

These quantities represent, respectively, the updated expected rework-arrival rate for line c and the updated probability that a rework job generated by that line requires complex processing.

[de: Are these posterior means going to be used later to select a routing policy in the second stage?]

[fq: Yes. For each realization of η, these posterior quantities characterize part of the updated predictive distribution of the future rework stream and therefore affect the evaluation of the candidate routing policies. The second-stage policy is selected according to

$$i^\star(\eta, z) \in \arg\max_i \mathbb{E}\left[Y(x_i, \xi) \mid \eta, z\right],$$

which accounts not only for these posterior means but also for the other operational random variables contained in ξ.]

The diagnostic measurements do not reveal the future arrival times or the complexity of individual rework jobs. Instead, they provide line-level information that updates the predictive distribution of the rework stream expected during the shift.

8.5 Routing policies

Let

$$\mathcal{X} = \{x_1, \ldots, x_k\}$$

be a finite set of candidate routing policies. Policy x_i is parameterized as

$$x_i = \left(a_v^i, a_q^i, \tau^i\right),$$

where a_v^i and a_q^i are weights associated with job value and queue congestion, respectively, and τ^i is a routing threshold. For an arriving rework job with value v and current workloads q^F and q^S at the fast and standard servers, respectively, policy x_i routes the job to the fast server if

$$a_v^i v - a_q^i \left(q^F - q^S\right) \geq \tau^i,$$

and to the standard server otherwise.

[de: Does "individual diagnostic signal" refer to the signal from inspecting individual jobs, like we had before? And does this signal still depend on the probing decision, z, like we had before? If not, then I will treat p_c^{hit} and p_c^{false} as constants.]

[fq: Sorry, my mistake. That came from the previous model, but I do not think we need it in this example, so I removed it. The vector η contains the pre-shift measurements of the assembly-line conditions and is used only to select one routing policy before the shift. Consequently, the parameters a_s^i, p_c^{hit}, and p_c^{false} should not be part of this example.]

[fq: I think we could reintroduce these parameters (if we want) by interpreting the quality-control inspection as also providing a binary indication of the likely complexity of each rework job. In that case, let $S_{c,j} = 1$ indicate that the jth rework job generated by line c is classified as likely to require complex processing. The actual complexity would still be represented by $D_{c,j}$, while

$$p_c^{\text{hit}} = \mathbb{P}(S_{c,j} = 1 \mid D_{c,j} = 1)$$

and

$$p_c^{\text{false}} = \mathbb{P}(S_{c,j} = 1 \mid D_{c,j} = 0)$$

would describe the accuracy of the quality-control signal.

After observing the pre-shift measurements η, we would select one routing policy for the entire shift. As each rework job arrives, the selected policy could use its quality-control signal $S_{c,j}$, together with its value and the current queue congestion, to decide between the fast and standard servers (something like $a_v^i v_{c,j} - a_q^i(q^F - q^S) + a_s^i S_{c,j} \geq \tau_i$.). This job-level signal would be distinct from the pre-shift measurement η_c. Therefore, I would treat p_c^{hit} and p_c^{false} as fixed parameters that do not depend on the probing decision z.]

After observing η, one policy is selected:

$$i^\star(\eta, z) \in \arg\max_{i=1,\ldots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \eta, z\right].$$

The selected policy remains fixed throughout the complete production shift. The pre-shift measurements affect the operation only through the identity of the selected policy; they do not enter directly into the routing rule applied to an individual rework job.

8.6 Decision structure

The model contains two optimization stages. The first-stage decision is the continuous probing vector

$$z = (z_1, \dots, z_m) \in [0,1]^m,$$

which determines how long each assembly line is stopped and diagnosed before the operating shift. This decision is made without observing the latent line conditions

$$\Theta = (\Theta_1, \dots, \Theta_m).$$

After choosing z, the diagnostic measurements

$$\eta = (\eta_1, \dots, \eta_m)$$

are observed, and the decision maker updates the distribution of Θ.

The second-stage decision is then the selection of one routing policy:

$$i^\star(\eta, z) \in \arg\max_{i=1,\dots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \eta, z\right].$$

The selected policy $x_{i^\star(\eta,z)}$ remains fixed throughout the entire operating shift. The individual routing decisions made as rework jobs arrive are applications of this fixed policy and do not constitute additional optimization stages.

Let

$$\xi = (\Theta, \mathcal{A}, \mathcal{D}, \mathcal{B})$$

denote the random elements that determine the operation of the rework system. Here, $\mathcal{A}$ contains the rework-arrival times and the corresponding lines of origin, $\mathcal{D}$ contains the rework-complexity, and $\mathcal{B}$ contains the potential service times.

We assume that, conditional on the latent line conditions Θ, the future operational random variables do not depend directly on the probing decision z or on the diagnostic measurements η. Moreover, the latent line conditions affect the future rework stream through the arrival processes and the complexity indicators, whereas the service times depend on the realized arrivals and complexities.

The dependence structure can be summarized as The sequence of events can be represented linearly as

$$\Theta \text{ is realized but remains unobserved} \quad z \longrightarrow \eta \longrightarrow x_{i^\star(\eta,z)} \longrightarrow (\mathcal{A}, \mathcal{D}, \mathcal{B}) \longrightarrow Y.$$

The resulting optimization problem is

$$\max_{z \in [0,1]^m} \left\{ -\alpha^\top z + \mathbb{E}_{\eta|z} \left[\max_{i=1,\dots,k} \mathbb{E}\left[Y(x_i, \xi) \mid \eta, z\right] \right] \right\}.$$

This formulation explicitly places the routing-policy selection after the information generated by probing has been observed. The pre-shift measurements affect the operation through the identity of the selected policy and do not enter directly into the routing rule applied during the shift.

Likelihood Ratio Derivation

[de: For Franco:] I started to work on the LR and score function, but got stuck on a few things that were not fully clear to me. Here are some notes though to get us started:

- η consists of the signals from each line: $(\eta_1, \eta_2, \ldots, \eta_C)$, where $C = |\mathcal{C}|$.

  [fq: Yes. Each η_c is a single measurement of the latent condition Θ_c of assembly line c.]

- ξ consists of $\Theta = (\Theta_1, \Theta_2, \ldots, \Theta_C)$, the arrival times of all jobs, the service times of all jobs (for both the standard and fast servers), the classes of the jobs, the complexities of the jobs, and the individual diagnostic signals.

  [fq: Yeah, this point was not clear at all. Now I want to write something like

  $$\xi = (\Theta, \mathcal{A}, \mathcal{D}, \mathcal{B}),$$

  where $\mathcal{A}$ contains the rework-arrival times and the corresponding lines of origin, $\mathcal{D}$ contains the rework-complexity, and $\mathcal{B}$ contains the potential service times at the fast and standard servers.

  At some point, I was mixing ideas from the previous example that do not make sense here. We now have neither separate job classes nor individual diagnostic signals in this example, because the line of origin provides the job class or type, which is already encoded in the arrival process, and probing is related to the assembly line rather than to the job itself.]

- Of the quantities in ξ, the ones that depend on z (via η) are Θ, the arrival times, and the complexities [fq: yes]. (This is assuming the diagnostic signals do not depend on z) [fq: yes, they do not depend on z and we do not longer have it in the example]. More precisely, the arrival times and the complexities of the jobs depend on Θ [fq: yes]. As a result, our likelihood will likely factorize into one term for Θ and another term for arrivals and complexities *given* Θ. This might look like

  $$f(\xi \mid \eta) = f(\Theta \mid \eta) \times f(\text{arrivals and complexities} \mid \Theta, \eta) \times f(\text{classes and diagnostic signals}).$$

  We can just ignore the last term because it will cancel out in the likelihood ratio. [fq: and we do not really have it now.]

  [fq: I think so, Here I am not quite sure. I want to say that the density factorizes as

  $$f(\xi \mid \eta, z) = f(\Theta \mid \eta, z) f(\mathcal{A} \mid \Theta) f(\mathcal{D} \mid \mathcal{A}, \Theta) f(\mathcal{B} \mid \mathcal{A}, \mathcal{D}).$$

  but I am not sure.]

- I'm still trying to figure out whether "arrivals" should be regarded as arrival times, interarrival times, or interarrival times for all jobs and Bernoulli's marking whether a job needed rework or not.

[fq: My first thought was to model separate arrival times for each assembly line, although I am not sure whether this has any particular advantage or disadvantage.

I do not think we need additional Bernoulli variables to indicate whether a produced unit requires rework. In this example, the quality-control step already determines whether a unit needs rework and, hence, every arrival in the model already corresponds to a job that needs to be processed.

We could include an additional random variable indicating whether each unit requires rework, and we would then also need to take an expectation over these outcomes, in addition to the expectation over η. Do you think including this could be useful?]

Notes on Softmax

Recall that our optimization problem is of the form

$$\max_{z\in[0,1]^m} \left\{-\alpha^\top z + q^\circ(z)\right\},$$

where

$$q^\circ(z) = \mathbb{E}_\eta \left[\max_{i=1,2,\ldots,k} \mathbb{E}\left[Y(x_i,\xi) \mid \eta\right]\right].$$

The resulting score-function estimator for the gradient of $q^\circ(z)$ will be

$$\widehat{\nabla_z q^\circ(z)} = \frac{1}{M}\sum_{j=1}^{M}\left[\frac{1}{n}\sum_{\ell=1}^{n} Y(x_{i^*(\eta_z^{(j)})};\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)}\nabla_z \log f(\xi_\ell;\eta_z^{(j)})\right], \qquad (1)$$

where $\eta_z^{(1)}, \eta_z^{(2)}, \ldots, \eta_z^{(M)}$ are the i.i.d. realizations of all probing variables used to estimate the outer expectation via crude Monte Carlo and

$$i^*(\eta_z^{(j)}) = \arg\max_{i=1,\ldots,k}\left[\frac{1}{n}\sum_{\ell=1}^{n} Y(x_i,\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)}\right],$$

i.e., the argmax of the likelihood-ratio-estimated value of $q^\circ(z)$ from only the jth outer replication.

A concern is that the estimation of the argmax $i^*(\eta_z^{(j)})$ will be highly variable, owing to the variability of the LR estimates of $q^\circ(z)$, which will in turn create potential instability in the LR/SF gradient estimator $\widehat{\nabla_z q^\circ(z)}$. One proposal to mitigate this to adopt ideas from the **softmax** to replace the gradient of the argmax with a convex combination of the gradients of the individual systems (x_i's), weighted by how close each x_i's LR estimated objective is to the optimal.

Let

$$Z_i(\eta_z^{(j)}) := \frac{1}{n}\sum_{\ell=1}^{n} Y(x_i,\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)} \quad \text{and} \quad Z^*(\eta_z^{(j)}) := \max_{i=1,\ldots,k} Z_i(\eta_z^{(j)}).$$

Define

$$\widehat{\nabla_z q^\circ(z)}^{\text{soft}} = \frac{1}{M}\sum_{j=1}^{M}\left[\sum_{i=1}^{k}\left[\frac{1}{n}\sum_{\ell=1}^{n} Y(x_i,\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)}\nabla_z \log f(\xi_\ell;\eta_z^{(j)})\right] w_i(\eta_z^{(j)})\right], \tag{2}$$

where

$$w_i(\eta_z^{(j)}) := \frac{\exp(-\beta(Z^*(\eta_z^{(j)}) - Z_i(\eta_z^{(j)}))}{\sum_{i=1}^{k}\exp(-\beta(Z^*(\eta_z^{(j)}) - Z_i(\eta_z^{(j)}))}$$

for some $\beta > 0$. The weights $w_i(\eta_z^{(j)})$ are highest for systems whose LR-estimated value of $q^\circ(z)$, $Z_i(\eta_z^{(j)})$, is close to the optimal $Z^*(\eta_z^{(j)})$ and rapidly decays to zero for inferior systems (x_i's). By construction, the weights sum to 1.

Notice that

$$\sum_{i=1}^{k}\left[\frac{1}{n}\sum_{\ell=1}^{n} Y(x_i,\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)}\nabla_z \log f(\xi_\ell;\eta_z^{(j)})\right] w_i(\eta_z^{(j)})$$

can be viewed as the expected value of a random gradient estimator that takes the value of

$$\frac{1}{n}\sum_{\ell=1}^{n} Y(x_i,\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)}\nabla_z \log f(\xi_\ell;\eta_z^{(j)})$$

with probability $w_i(\eta_z^{(j)})$ for $i = 1, 2, \ldots, k$.

Notice that as $\beta \to \infty$,

$$\widehat{\nabla_z q^\circ(z)}^{\text{soft}} \to \widehat{\nabla_z q^\circ(z)}$$

since $w_i(\eta_z^{(j)}) \to 1$ for $i = i^*(\eta_z^{(j)})$ and $w_i(\eta_z^{(j)}) \to 0$ for all $i \neq i^*(\eta_z^{(j)})$. This might suggest that later in a search we should increase β, or that we can make theoretical statements about how the search performs as $\beta \to \infty$.

New Notes on Incorporating Uncertainty

The softmax idea above attempts to mitigate issues with mis-estimating the argmax by mixing in the gradients of other terms that could have been the argmax, weighted by how far their objective estimates are from the max objective estimate. This approach does not directly account for the variability in the LR estimates, which may vary across x's. This variability is important, because it represents how likely the argmax could be something different.

Recall that the LR objective estimate for x_i is

$$Z_i(\eta_z^{(j)}) = \frac{1}{n}\sum_{\ell=1}^{n} Y(x_i,\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)}.$$

For each x_i, we'd like to estimate $p_i(\eta_z^{(j)}) := P(Z_i(\eta_z^{(j)}) \geq Z_{i'}(\eta_z^{(j)})$ for all $i' \neq i$), i.e., the probability that system x_i is the argmax (based on the LR estimators)

in our setting. There are several approaches we could take to estimate p_i (coming soon), but what we are ultimately working toward is a gradient estimator

$$\widehat{\nabla_z q^\circ(z)}^{\text{psoft}} = \frac{1}{M}\sum_{j=1}^{M}\left[\sum_{i=1}^{k}\left[\frac{1}{n}\sum_{\ell=1}^{n} Y(x_i,\xi_\ell)\frac{f(\xi_\ell;\eta_z^{(j)})}{f(\xi_\ell)}\nabla_z \log f(\xi_\ell;\eta_z^{(j)})\right]\hat{p}_i(\eta_z^{(j)})\right], \tag{3}$$

where $\hat{p}_i(\eta_z^{(j)})$ is an estimator of $p_i(\eta_z^{(j)})$. Thus, instead of softening based on the "optimality gaps" (as in the softmax), we soften based on the probability that each x_i is the argmax. As we become more certain of the true argmax, e.g., as $n \to \infty$, we'll have $p_{i^*(\eta_z^{(j)})}(\eta_z^{(j)}) \to 1$ and $p_{i'}(\eta_z^{(j)}) \to 0$ for all $i' \neq i^*(\eta_z^{(j)})$, hence we recover the original estimator asymptotically.

Two approaches that come to mind for estimating $p_i(\eta_z^{(j)})$:

- Model the $Z_i(\eta_z^{(j)})$s as a multivariate normal distribution. The mean vector is just the vector of the realized $Z_i(\eta_z^{(j)})$s and the covariance matrix is calculated from looking at the n pairings of the terms from any two $Z_i(\eta_z^{(j)})$ and $Z_{i'}(\eta_z^{(j)})$, paired by the index ℓ. When k is not too large (< 25), it should be easy enough to compute the CDF value for the MVN.
- Use bootstrapping. For each bootstrap instance, draw n values of the ℓ's (with replacement), compute the resulting $Z_i^\star(\eta_z^{(j)})$—where * means bootstrapped, and record the argmax of the $Z_i^\star(\eta_z^{(j)})$s. Repeat so many times and use the sample proportions that each system is the argmax as the $\hat{p}$s.